

Which graphics card is the best? The answer is simple: one that gives you optimum performance at the price you can afford. We help you choose the right card whether you are looking for killer effects or plain vanilla gaming

Video cards cost a bundle. The set of applications that one typically runs does not call for an investment into such hardware. Motherboards are now commonly equipped with an onboard graphics chipset that does an adequate job of satiating the needs of working with an office suite, exchanging e-mail, browsing the Internet, even enjoying the odd game or two. You will thus find that unless you specifically ask otherwise, your local PC assembler will consider it normal, supplying you with a system with an onboard video chipset. The computer market, however, is becoming more attractive and affordable to the masses; this fact is also reflected in the falling prices of video cards, or rather the availability of well performing, pocket-able solutions. For a little over Rs 1,000, you can find yourself a decent piece of hardware that will dedicate itself to painting your monitor with pixels.

But why would you need to invest in such a solution? If you wish to entertain yourself with DVD movies, you need a current breed of video cards that carry hardware decoders that will take the stress off your processor; if you find yourself indulging in more than a weekly session of intense 3D-gaming, well, nothing beats a good frag at 60 frames per second (fps)—something only an AGP card can offer with elan. Even professionals who find their source of income in 3dsmax or a CAD application will be more comfortable powering their 21-inch monitor with something that supports a resolution greater than 1280x1024 with flicker-free refresh rates that exceed 75 Hz. There are cards that support dual-monitors (one of those luxuries that soon becomes a necessity)—the Matrox Parhelia can run three! There are those that can output to a television; talk with your camcorder, exchanging movies; bring you cable television and double as a video recorder, saving your favourite programs on your hard drive, to be enjoyed at your leisure. Applications such as these further emphasise the role of your personal computer as the hub of digital information-exchange and entertainment.

Of all the applications, games come across as a determinant of performance, being at the top of a food chain that constitutes power-hungry applications—if your hardware can run a game, other tasks become a walk in the park. You will thus find that even the lowliest of cards meant for gaming, adequately run CAD applications, or drive a large monitor at a comfortable refresh rate and resolution. As you climb further up the price ladder, you find these cards with features that let you perform the tasks listed above. Thus, at the very top you will find a card similar to the ATI ALL-IN-WONDER 9700 PRO, one that is built around a gaming chipset and offers a TV Tuner, a remote control, the ability to record TV programs or to connect external hardware such as a flat-panel or a camcorder. While the low-end cards are reasonably priced, beginning around Rs 1,000, the high-end monsters exact a price in excess of Rs 20,000. With this in mind, we have tested 35 cards from the leading vendors in the Indian market, evaluating each one for two application areas: casual gaming and high-end gaming. Let us help you choose the right video card for your needs.

How we tested

Having established the areas of casual gaming and high end gaming on which to evaluate a card, we then defined the areas according to certain resolutions and feature sets. For casual gaming the score split was 60 per cent performance and 40 per cent price, while for high-end gaming the split was 80 per cent performance and 20 per cent price. A card that would perform with satisfaction at 640x480 and 800x600, while being affordable (not greater than Rs 5,000) would be considered ideal for casual gaming. For high-end gaming, we decided to weigh a card based on its performance at resolutions of 1024x768 and 1280x1024, with the logic that cards that cost a bundle should perform accordingly. Furthermore, features such as dual-monitor support, bundled VR glasses, a TV Tuner chip, API compatibility, etc also contributed to the scores of a card for the high-end gaming area.

We tested all the cards on the same set of benchmarks, but awarded them differently based on the resolutions run. With the benchmark scores in hand, we then calculated a Suitability To Task Index (STTI), which was an amalgamation of pure performance and price. We thus have an STTI score for the application area of casual gaming (the higher a card scores, the better suited it is to this task) and an STTI score for high-end gaming (again, the greater a card scores, the more suited it is to the area of high-end gaming). As an example, a card may score 70 STTI for casual gaming and 40 STTI for high-end gaming, making it an ideal buy for the casual gamer but an avoidable investment for someone who demands greater performance at high resolutions, advanced features such as DirectX 8.1/9.0 compliance, etc.

The benchmarks chosen were run at resolutions of 640x480, 800x600, 1024x768 and 1280x1024. Except for 3DMark2001 SE, which was run at 800x600 and 1024x768 resolution. What follows is a list of benchmarks used and the reason for using them. You will notice that all tests, synthetic or real-world, were related to games—if a card copes well with these set of hurdles, you can be sure that any video application you throw at it will be tackled with grace.

Quake III Arena: The venerable OpenGL-based engine still remains an intensive test, especially for cards with older chipsets. The ability of the game engine to scale higher or lower according to the hardware allows us to evaluate the performance of the graphics cards by keeping the rest of the test system the same. Moreover, most of the favourites in the Indian gaming market are derivatives of the *Quake III Arena* engine. The age of *Quake III Arena*, however, led us to delegate more importance to it in the casual gaming area—it got the highest weightage under this sector. For a card's performance under OpenGL for the high-end gaming area we used a newer game, noted below.

